
Data Cleaning & Preliminary Analysis

EMIF Group Project 1

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Data Cleaning : Weekly Frequency

Selected Variables

- **WTI** : WTI crude oil futures (CL1 Comdty)
- **HY** : Bloomberg US Corporate High Yield Index (LF98YW Index)
- **SP500** : S&P 500 Index (SPX Index)
- **US2Y** : US Government 2-Year Yield (USGG2YR Index)

1. Data Loading

```
In [36]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
from scipy import stats
import statsmodels as sm
from statsmodels.tsa.api import VAR
from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.graphics.tsaplots import plot_acf
```

```
In [37]: raw_path = Path("../data/raw/data_hec_projet_1.xlsx")
daily_raw = pd.read_excel(raw_path, sheet_name="Daily", skiprows=5, index

print("Raw dimensions:", daily_raw.shape)
daily_raw.head()
```

Raw dimensions: (9443, 13)

Out[37]:

	PX_LAST	PX_LAST.1	PX_LAST.2	PX_LAST.3	PX_LAST.4	PX_LAST.5	PX_LAST.6
Dates							
1990-01-01	21.82	19.69	77.8405	NaN	NaN	353.40	567.34
1990-01-02	22.89	19.90	81.0621	NaN	NaN	359.69	568.96
1990-01-03	23.68	20.95	83.0504	NaN	NaN	358.76	569.10
1990-01-04	23.41	20.78	81.1315	NaN	NaN	355.67	571.01
1990-01-05	23.08	20.75	78.7509	NaN	NaN	352.20	566.36

```
In [38]: #filtering for selected variables
df_daily = daily_raw.iloc[:, [0, 5, 10, 11]].copy()
df_daily.columns = ["WTI", "SP500", "US2Y", "HY"]

df_daily.head()
```

Out[38]:

	WTI	SP500	US2Y	HY
Dates				
1990-01-01	21.82	353.40	7.841	16.08
1990-01-02	22.89	359.69	7.875	16.08
1990-01-03	23.68	358.76	7.927	16.08
1990-01-04	23.41	355.67	7.910	16.08
1990-01-05	23.08	352.20	7.885	16.08

```
In [39]: #identify when HY frequency changes from monthly to higher frequency
hy_changes = df_daily["HY"].dropna()
hy_changes = hy_changes[hy_changes != hy_changes.shift(1)]

gaps = hy_changes.index.to_series().diff().dt.days.dropna()

print(gaps[gaps >= 20].index[-1].date())

1998-07-31
```

```
In [40]: #restrict sample to period with daily HY frequency
df_daily = df_daily.loc["1998-08-01":]
```

2. Weekly resampling of daily data

```
In [41]: #resample to weekly frequency (last observation of the week)
df_weekly = df_daily.resample("W-FRI").last()

print(f"Period: {df_weekly.index[0].date()} to {df_weekly.index[-1].date()}")
df_weekly.head()
```

Period: 1998-08-07 to 2026-03-13

```
Out[41]:          WTI    SP500    US2Y    HY
```

Dates				
1998-08-07	14.14	1089.45	5.328	9.03
1998-08-14	13.64	1062.75	5.318	9.31
1998-08-21	13.37	1081.24	5.197	9.56
1998-08-28	13.50	1027.14	4.905	10.01
1998-09-04	14.59	973.89	4.902	10.56

3. Returns to Excess Returns

```
In [42]: #construct return-based variables for VAR

df_ret = df_weekly.copy()

#log-returns (WTI, SP500)
for col in ["WTI", "SP500"]:
    df_ret[col] = np.log(df_weekly[col] / df_weekly[col].shift(1))

#weekly risk-free rate (decimal)
df_ret["US2Y"] = df_weekly["US2Y"] / (100 * 52)

#VAR 1 variables
ret_wti = df_ret["WTI"].rename("ret_wti")
er_sp500 = (df_ret["SP500"] - df_ret["US2Y"]).rename("er_sp500")

#change in HY spread (decimal)
spread_hy = df_weekly["HY"] - df_weekly["US2Y"]
d_spread_hy = ((spread_hy - spread_hy.shift(1)) / 100).rename("d_spread_hy")
```

4. Realized Variance

```
In [43]: #construct realized variance series for VAR

#daily log-returns (WTI, SP500)
daily_ret_wti    = np.log(df_daily["WTI"] / df_daily["WTI"].shift(1)).drop
daily_ret_sp500  = np.log(df_daily["SP500"] / df_daily["SP500"].shift(1)).

#daily change in HY spread
daily_spread     = df_daily["HY"] - df_daily["US2Y"]
daily_d_spread   = ((daily_spread - daily_spread.shift(1)) / 100).dropna()

#weekly realized variance
rv_wti          = (daily_ret_wti ** 2).resample("W-FRI").sum().rename("rv_wti")
rv_sp500        = (daily_ret_sp500 ** 2).resample("W-FRI").sum().rename("rv_sp500")
rv_hy           = (daily_d_spread ** 2).resample("W-FRI").sum().rename("rv_hy")
```

Weekly realized variance is computed as the sum of squared daily returns:

$$RV_t = \sum_{d \in \text{week } t} r_d^2$$

where r_d denotes daily log-returns (or daily changes in the HY spread).

```
In [44]: #cap extreme values in realized variance (upper tail)

rv_wti    = rv_wti.clip(upper=rv_wti.quantile(0.99))
rv_sp500   = rv_sp500.clip(upper=rv_sp500.quantile(0.99))
rv_hy      = rv_hy.clip(upper=rv_hy.quantile(0.99))

pd.concat([rv_wti, rv_sp500, rv_hy], axis=1).describe().round(6)
```

```
Out[44]:
```

	rv_wti	rv_sp500	rv_hy
count	1441.000000	1441.000000	1441.000000
mean	0.002575	0.000654	0.000005
std	0.003219	0.001052	0.000011
min	0.000007	0.000003	0.000000
25%	0.000853	0.000137	0.000001
50%	0.001595	0.000309	0.000002
75%	0.002932	0.000713	0.000004
max	0.019726	0.007425	0.000091

5. Final Cleaning & Export

```
In [45]: #VAR 1 variables
var1 = pd.concat([ret_wti, er_sp500, d_spread_hy], axis=1).dropna()

#VAR 2 variables: realized variance from daily data
var2 = pd.concat([rv_wti, rv_sp500, rv_hy], axis=1)

#combine datasets
df_final = pd.concat([var1, var2], axis=1, sort=False).dropna()
df_final = df_final.asfreq('W-FRI')

print(f"Period: {df_final.index[0].date()} to {df_final.index[-1].date()}")
print(f"Observations: {len(df_final)}")
df_final.head()
```

Period: 1998-08-14 to 2026-03-13

Observations: 1440

```
Out[45]:
```

	ret_wti	er_sp500	d_spread_hy	rv_wti	rv_sp500	rv_hy
Dates						
1998-08-14	-0.036001	-0.025836	0.00290	0.004282	0.000611	0.000003
1998-08-21	-0.019993	0.016249	0.00371	0.002764	0.000771	0.000006
1998-08-28	0.009676	-0.052274	0.00742	0.001773	0.001876	0.000022
1998-09-04	0.077647	-0.054178	0.00553	0.006006	0.006555	0.000033
1998-09-11	-0.002058	0.034574	0.00222	0.001817	0.004283	0.000004

```
In [46]: #summary statistics of final dataset
print(df_final.describe().round(6).to_string())

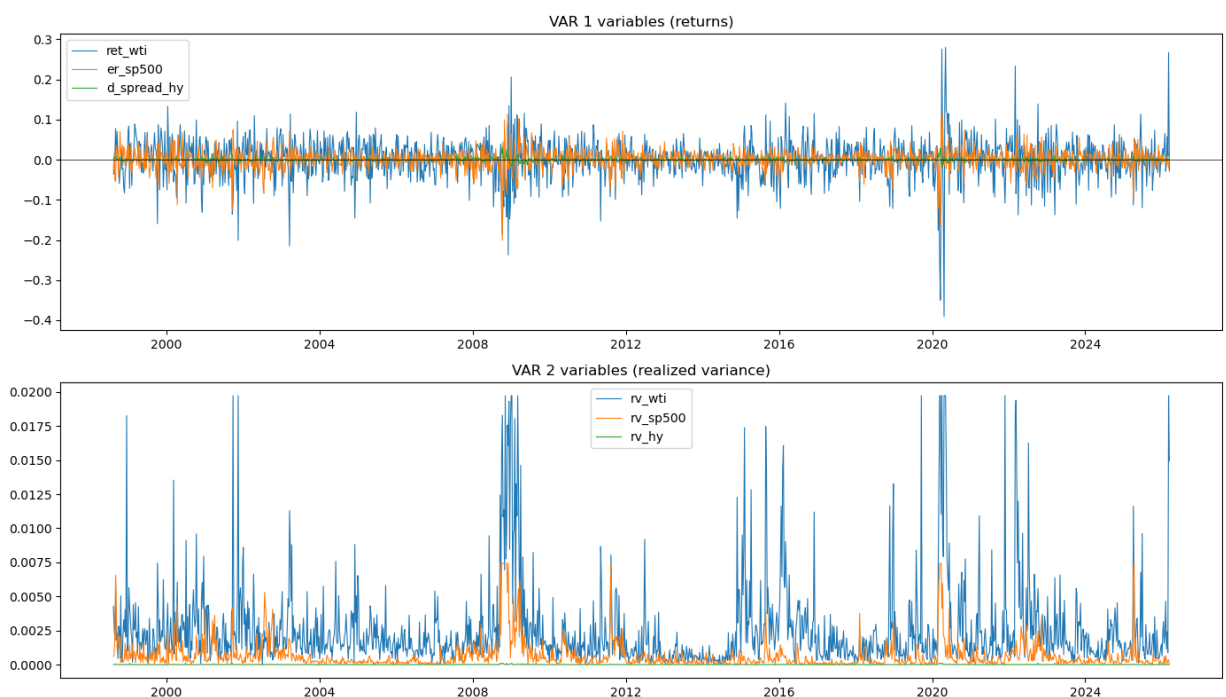
#plot VAR variables
fig, axes = plt.subplots(2, 1, figsize=(14, 8))

for col in ["ret_wti", "er_sp500", "d_spread_hy"]:
    axes[0].plot(df_final.index, df_final[col], label=col, linewidth=0.8)
axes[0].axhline(0, color="black", linewidth=0.5)
axes[0].set_title("VAR 1 variables (returns)")
axes[0].legend()

for col in ["rv_wti", "rv_sp500", "rv_hy"]:
    axes[1].plot(df_final.index, df_final[col], label=col, linewidth=0.8)
axes[1].set_title("VAR 2 variables (realized variance)")
axes[1].legend()

plt.tight_layout()
plt.show()
```

	ret_wti	er_sp500	d_spread_hy	rv_wti	rv_sp500	
rv_hy						
count	1440.000000	1440.000000	1440.000000	1440.000000	1440.000000	1
mean	0.001246	0.000807	-0.000003	0.002577	0.000654	
std	0.051711	0.024900	0.003039	0.003220	0.001052	
min	-0.390397	-0.201152	-0.015265	0.000007	0.000003	
25%	-0.024628	-0.011246	-0.001304	0.000853	0.000137	
50%	0.004566	0.002304	-0.000215	0.001596	0.000309	
75%	0.031516	0.014116	0.001122	0.002932	0.000712	
max	0.279942	0.114193	0.031480	0.019726	0.007425	



```
In [47]: df_final.to_csv(Path("../data/processed/weekly_cleaned_data.csv"))
print("Exported: ../data/processed/weekly_cleaned_data.csv")
```

Exported: ../data/processed/weekly_cleaned_data.csv

6. Preliminary Statistical Analysis

```
In [48]: #summary statistics and normality tests for returns

daily_returns = pd.concat([daily_ret_wti, daily_ret_sp500, daily_d_sprea
weekly_returns = pd.concat([ret_wti, er_sp500, d_spread_hy], axis=1).drop

def moment_table(df):
    return pd.DataFrame({
        "mean": df.mean(),
        "std": df.std(),
        "skewness": df.skew(),
        "kurtosis": df.kurtosis() + 3,
        "excess_kurtosis": df.kurtosis(),
    })

def jb_pvalues(ret):
    return pd.Series({c: stats.jarque_bera(ret[c].dropna().values).pvalue

mom_daily = moment_table(daily_returns)
mom_weekly = moment_table(weekly_returns).add_prefix("W_")

compare_moments = pd.concat([
    mom_daily[["skewness", "kurtosis", "excess_kurtosis"]].reset_index(dr
    mom_weekly[["W_skewness", "W_kurtosis", "W_excess_kurtosis"]].reset_i
], axis=1)
compare_moments.index = ["ret_wti", "er_sp500", "d_spread_hy"]

display(compare_moments)

jb_daily = jb_pvalues(daily_returns).rename("JB_p_daily")
jb_weekly = jb_pvalues(weekly_returns).rename("JB_p_weekly")

compare_jb = pd.concat([jb_daily.reset_index(drop=True), jb_weekly.reset_
compare_jb.index = ["ret_wti", "er_sp500", "d_spread_hy"]

display(compare_jb)
```

	skewness	kurtosis	excess_kurtosis	W_skewness	W_kurtosis	W_exces
ret_wti	-1.911451	50.736005	47.736005	-0.602216	9.370869	
er_sp500	-0.348547	13.603476	10.603476	-0.820673	9.637627	
d_spread_hy	0.540489	97.640223	94.640223	1.956947	23.881764	

	JB_p_daily	JB_p_weekly
ret_wti	0.0	0.0
er_sp500	0.0	0.0
d_spread_hy	0.0	0.0

```

In [49]: #autocorrelation check

lags_to_test = [4, 8, 12]

#Ljung-Box on weekly returns (EMH)
lb_ret = {}
for col in weekly_returns.columns:
    test = acorr_ljungbox(weekly_returns[col].dropna(), lags=lags_to_test)
    lb_ret[col] = test[["lb_stat", "lb_pvalue"]]

display(pd.concat(lb_ret, axis=0).round(4))

#Ljung-Box on realized variance (volatility clustering)
weekly_variances = pd.concat([rv_wti, rv_sp500, rv_hy], axis=1).dropna()

lb_var = {}
for col in weekly_variances.columns:
    test = acorr_ljungbox(weekly_variances[col].dropna(), lags=lags_to_test)
    lb_var[col] = test[["lb_stat", "lb_pvalue"]]

display(pd.concat(lb_var, axis=0).round(4))

```

		lb_stat	lb_pvalue
ret_wti	4	6.4865	0.1656
	8	18.4865	0.0179
	12	24.0645	0.0199
er_sp500	4	18.0847	0.0012
	8	24.6214	0.0018
	12	29.3075	0.0035
d_spread_hy	4	156.8405	0.0000
	8	167.8667	0.0000
	12	175.8268	0.0000

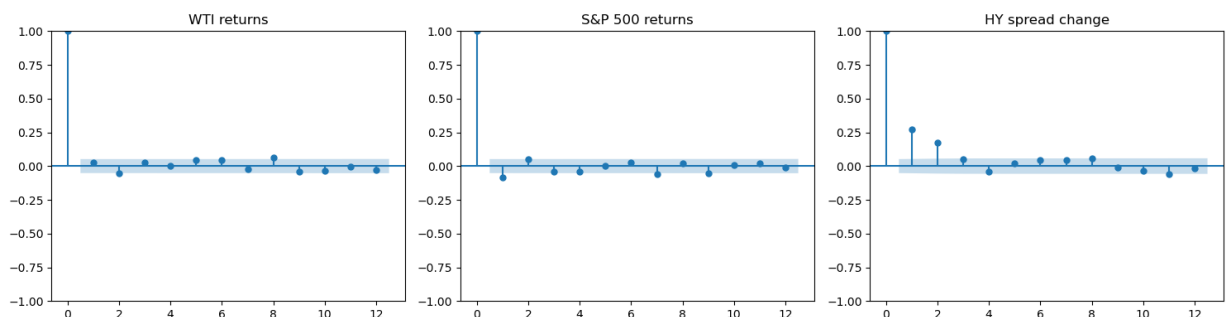
		lb_stat	lb_pvalue
rv_wti	4	1165.8994	0.0
	8	1761.5447	0.0
	12	2010.8053	0.0
rv_sp500	4	2092.4375	0.0
	8	2844.4460	0.0
	12	3205.9712	0.0
rv_hy	4	766.2628	0.0
	8	1004.9867	0.0
	12	1222.4130	0.0

```
In [50]: #visual inspection of autocorrelation (returns)

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

plot_acf(ret_wti.dropna(), lags=12, ax=axes[0], title="WTI returns")
plot_acf(er_sp500.dropna(), lags=12, ax=axes[1], title="S&P 500 returns")
plot_acf(d_spread_hy.dropna(), lags=12, ax=axes[2], title="HY spread chan

plt.tight_layout()
plt.show()
```

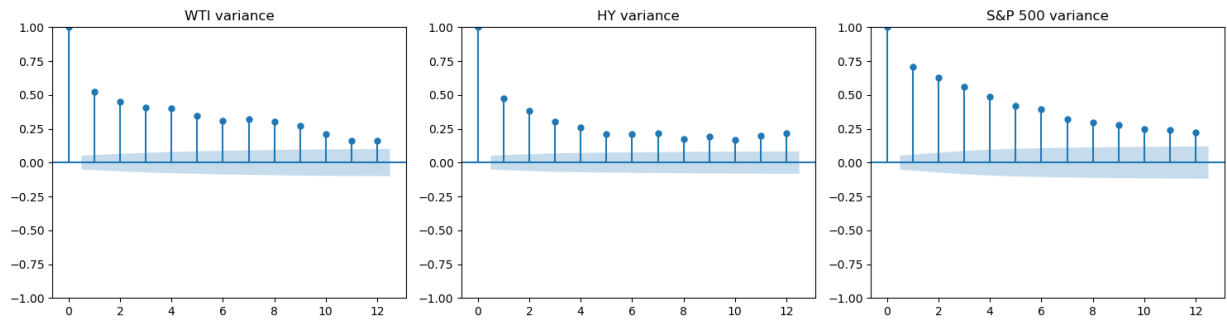


```
In [51]: #visual inspection of autocorrelation (realized variance)

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

plot_acf(rv_wti.dropna(), lags=12, ax=axes[0], title="WTI variance")
plot_acf(rv_hy.dropna(), lags=12, ax=axes[1], title="HY variance")
plot_acf(rv_sp500.dropna(), lags=12, ax=axes[2], title="S&P 500 variance")

plt.tight_layout()
plt.show()
```



VAR Models

Testing for Stationarity

```
In [52]: #ADF test for stationarity

variables = df_final.columns.tolist()
for i in variables:
    result = sm.tsa.stattools.adfuller(df_final[i])
    print(f"{i:<12}: ADF = {result[0]:8.4f}, nb of lags used {result[2]:>}
```

ret_wti	:	ADF = -11.8303, nb of lags used 8, p-value = 0.000000
er_sp500	:	ADF = -20.5501, nb of lags used 3, p-value = 0.000000
d_spread_hy	:	ADF = -7.9360, nb of lags used 22, p-value = 0.000000
rv_wti	:	ADF = -7.4003, nb of lags used 10, p-value = 0.000000
rv_sp500	:	ADF = -8.5132, nb of lags used 6, p-value = 0.000000
rv_hy	:	ADF = -6.3951, nb of lags used 11, p-value = 0.000000

Number of lags selection

```
In [53]: #lag selection for VAR 1 (returns)

var1_data = df_final[["ret_wti", "d_spread_hy", "er_sp500"]]
modell1 = VAR(var1_data)
results1 = modell1.select_order(maxlags=20)
print(results1.summary())
```

VAR Order Selection (* highlights the minimums)

	AIC	BIC	FPE	HQIC
0	-25.49	-25.48	8.527e-12	-25.48
1	-25.64	-25.59*	7.330e-12	-25.62
2	-25.66	-25.58	7.198e-12	-25.63*
3	-25.66	-25.55	7.197e-12	-25.62
4	-25.66	-25.52	7.149e-12	-25.61
5	-25.67	-25.49	7.103e-12	-25.60
6	-25.67	-25.46	7.108e-12	-25.59
7	-25.68	-25.43	7.064e-12	-25.58
8	-25.68	-25.40	7.032e-12	-25.58
9	-25.68	-25.37	7.030e-12	-25.56
10	-25.68	-25.33	7.058e-12	-25.55
11	-25.68*	-25.30	7.020e-12*	-25.54
12	-25.68	-25.27	7.062e-12	-25.52
13	-25.67	-25.23	7.073e-12	-25.51
14	-25.67	-25.20	7.086e-12	-25.49
15	-25.67	-25.16	7.134e-12	-25.48
16	-25.66	-25.11	7.193e-12	-25.45
17	-25.65	-25.07	7.259e-12	-25.43
18	-25.65	-25.04	7.224e-12	-25.43
19	-25.66	-25.02	7.153e-12	-25.42
20	-25.66	-24.98	7.184e-12	-25.41

In [54]: *#lag selection for VAR 2 (realized variances)*

```
var2_data = df_final[["rv_wti", "rv_hy", "rv_sp500"]]
model2 = VAR(var2_data)
results2 = model2.select_order(maxlags=20)
print(results2.summary())
```

VAR Order Selection (* highlights the minimums)

	AIC	BIC	FPE	HQIC
0	-48.79	-48.78	6.437e-22	-48.79
1	-49.78	-49.74	2.404e-22	-49.76
2	-49.89	-49.81*	2.152e-22	-49.86
3	-49.91	-49.79	2.120e-22	-49.86
4	-49.92	-49.78	2.085e-22	-49.87*
5	-49.92	-49.74	2.088e-22	-49.85
6	-49.92	-49.71	2.085e-22	-49.84
7	-49.93	-49.68	2.072e-22	-49.84
8	-49.93	-49.65	2.075e-22	-49.82
9	-49.92	-49.61	2.082e-22	-49.81
10	-49.93	-49.58	2.077e-22	-49.80
11	-49.93	-49.55	2.075e-22	-49.79
12	-49.93	-49.52	2.065e-22	-49.78
13	-49.93*	-49.49	2.060e-22*	-49.77
14	-49.93	-49.45	2.070e-22	-49.75
15	-49.93	-49.42	2.071e-22	-49.74
16	-49.92	-49.38	2.085e-22	-49.72
17	-49.92	-49.35	2.080e-22	-49.71
18	-49.91	-49.30	2.102e-22	-49.69
19	-49.91	-49.27	2.109e-22	-49.67
20	-49.90	-49.23	2.124e-22	-49.65

VAR 1 — Returns

In [55]: *#estimate VAR(2) for returns*

```
var1_fitted = model1.fit(2)
print(var1_fitted.summary())
```

```
Summary of Regression Results
=====
Model:                                VAR
Method:                               OLS
Date:                                Sat, 04, Apr, 2026
Time:                                22:14:04

-----
No. of Equations:                    3.00000    BIC:                                -25.5605
Nobs:                                1438.00    HQIC:                               -25.6087
Log likelihood:                      12333.1    FPE:                                7.34145e-12
AIC:                                 -25.6375    Det(Omega_mle):                     7.23527e-12

-----
Results for equation ret_wti
=====
=====
              coefficient          std. error          t-stat
prob
-----
const              0.001519          0.001356          1.120
0.263
```

L1.ret_wti	-0.003263	0.027648	-0.118
0.906			
L1.d_spread_hy	-2.661819	0.613157	-4.341
0.000			
L1.er_sp500	-0.138745	0.071476	-1.941
0.052			
L2.ret_wti	-0.081287	0.027898	-2.914
0.004			
L2.d_spread_hy	-0.706340	0.586042	-1.205
0.228			
L2.er_sp500	-0.046843	0.069516	-0.674
0.500			

=====

=====

Results for equation d_spread_hy

=====

=====

	coefficient	std. error	t-stat
prob			

const	0.000006	0.000077	0.077
0.939			
L1.ret_wti	-0.001269	0.001564	-0.812
0.417			
L1.d_spread_hy	0.175673	0.034682	5.065
0.000			
L1.er_sp500	-0.010658	0.004043	-2.636
0.008			
L2.ret_wti	0.002658	0.001578	1.684
0.092			
L2.d_spread_hy	0.112009	0.033149	3.379
0.001			
L2.er_sp500	-0.005937	0.003932	-1.510
0.131			

=====

=====

Results for equation er_sp500

=====

=====

	coefficient	std. error	t-stat
prob			

const	0.000900	0.000656	1.372
0.170			
L1.ret_wti	-0.005093	0.013379	-0.381
0.703			
L1.d_spread_hy	-0.359169	0.296702	-1.211
0.226			
L1.er_sp500	-0.103670	0.034587	-2.997
0.003			
L2.ret_wti	-0.021701	0.013500	-1.608
0.108			
L2.d_spread_hy	-0.107814	0.283581	-0.380

```

0.704
L2.er_sp500          0.034718          0.033638          1.032
0.302
=====
=====

```

Correlation matrix of residuals

```

          ret_wti  d_spread_hy  er_sp500
ret_wti      1.000000    -0.290130  0.251283
d_spread_hy  -0.290130     1.000000 -0.644852
er_sp500      0.251283    -0.644852  1.000000

```

VAR 2 — Realized Variances

In [56]: *#estimate VAR(4) for realized variances*

```

var2_fitted = model2.fit(4)
print(var2_fitted.summary())

```

```

Summary of Regression Results
=====
Model:                                VAR
Method:                               OLS
Date:                Sat, 04, Apr, 2026
Time:                22:14:04
-----
No. of Equations:      3.00000    BIC:                -49.7474
Nobs:                  1436.00    HQIC:               -49.8371
Log likelihood:        29747.6    FPE:                2.15185e-22
AIC:                   -49.8905    Det(Omega_mle):     2.09445e-22
-----
Results for equation rv_wti
=====
=====

```

	coefficient	std. error	t-stat
prob			

const	0.000610	0.000097	6.298
0.000			
L1.rv_wti	0.270146	0.027358	9.875
0.000			
L1.rv_hy	36.369729	7.895391	4.606
0.000			
L1.rv_sp500	0.348925	0.104309	3.345
0.001			
L2.rv_wti	0.129095	0.028498	4.530
0.000			
L2.rv_hy	-12.821212	8.101662	-1.583
0.114			
L2.rv_sp500	-0.007210	0.112333	-0.064
0.949			

L3.rv_wti	0.063242	0.028561	2.214
0.027			
L3.rv_hy	15.377176	8.132718	1.891
0.059			
L3.rv_sp500	-0.050212	0.111742	-0.449
0.653			
L4.rv_wti	0.128501	0.027477	4.677
0.000			
L4.rv_hy	-13.353687	7.986744	-1.672
0.095			
L4.rv_sp500	0.201537	0.103167	1.954
0.051			

=====

=====

Results for equation rv_hy

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	coefficient	std. error	t-stat
prob			

const	0.000000	0.000000	0.564
0.573			
L1.rv_wti	0.000271	0.000101	2.688
0.007			
L1.rv_hy	0.249394	0.029041	8.588
0.000			
L1.rv_sp500	0.001400	0.000384	3.650
0.000			
L2.rv_wti	0.000097	0.000105	0.927
0.354			
L2.rv_hy	0.083933	0.029799	2.817
0.005			
L2.rv_sp500	0.000732	0.000413	1.773
0.076			
L3.rv_wti	-0.000194	0.000105	-1.848
0.065			
L3.rv_hy	-0.007689	0.029913	-0.257
0.797			
L3.rv_sp500	0.000645	0.000411	1.569
0.117			
L4.rv_wti	0.000389	0.000101	3.848
0.000			
L4.rv_hy	-0.028015	0.029377	-0.954
0.340			
L4.rv_sp500	0.000036	0.000379	0.095
0.924			

=====

=====

Results for equation rv_sp500

=====

=====

	coefficient	std. error	t-stat
prob			

```

-----
const          0.000117          0.000027          4.391
0.000
L1.rv_wti      0.017731          0.007527          2.356
0.018
L1.rv_hy       6.335052          2.172234          2.916
0.004
L1.rv_sp500    0.435047          0.028698          15.159
0.000
L2.rv_wti     -0.017863          0.007841          -2.278
0.023
L2.rv_hy       6.215353          2.228985          2.788
0.005
L2.rv_sp500    0.183382          0.030906          5.934
0.000
L3.rv_wti     -0.004340          0.007858          -0.552
0.581
L3.rv_hy       1.067911          2.237529          0.477
0.633
L3.rv_sp500    0.088435          0.030743          2.877
0.004
L4.rv_wti      0.003636          0.007560          0.481
0.631
L4.rv_hy      -0.772435          2.197368          -0.352
0.725
L4.rv_sp500    0.011140          0.028384          0.392
0.695
=====
=====

```

Correlation matrix of residuals

	rv_wti	rv_hy	rv_sp500
rv_wti	1.000000	0.213387	0.225887
rv_hy	0.213387	1.000000	0.395828
rv_sp500	0.225887	0.395828	1.000000

Granger Causality Tests

```
In [57]: def granger_table(fitted, pairs):
    rows = []
    for causing, caused in pairs:
        res = fitted.test_causality(caused=caused, causing=causing)
        rows.append({
            "Causing variable": causing,
            "Caused variable": caused,
            "Test statistic": round(res.test_statistic, 4),
            "p-value": round(res.pvalue, 4),
            "Significant at 5%": "Yes" if res.pvalue < 0.05 else "No",
        })
    return pd.DataFrame(rows)

pairs_var1 = [
    ("ret_wti", "er_sp500"),
    ("ret_wti", "d_spread_hy"),
    ("er_sp500", "ret_wti"),
    ("er_sp500", "d_spread_hy"),
    ("d_spread_hy", "ret_wti"),
    ("d_spread_hy", "er_sp500"),
]

pairs_var2 = [
    ("rv_wti", "rv_sp500"),
    ("rv_wti", "rv_hy"),
    ("rv_sp500", "rv_wti"),
    ("rv_sp500", "rv_hy"),
    ("rv_hy", "rv_wti"),
    ("rv_hy", "rv_sp500"),
]

print("Granger Causality: VAR 1 (returns)")
display(granger_table(var1_fitted, pairs_var1))

print("\nGranger Causality: VAR 2 (realized variances)")
display(granger_table(var2_fitted, pairs_var2))
```

Granger Causality: VAR 1 (returns)

	Causing variable	Caused variable	Test statistic	p-value	Significant at 5%
0	ret_wti	er_sp500	1.3625	0.2561	No
1	ret_wti	d_spread_hy	1.7520	0.1735	No
2	er_sp500	ret_wti	1.9653	0.1402	No
3	er_sp500	d_spread_hy	4.1346	0.0161	Yes
4	d_spread_hy	ret_wti	11.5768	0.0000	Yes
5	d_spread_hy	er_sp500	0.9272	0.3957	No

Granger Causality: VAR 2 (realized variances)

	Causing variable	Caused variable	Test statistic	p-value	Significant at 5%
0	rv_wti	rv_sp500	2.2974	0.0567	No
1	rv_wti	rv_hy	7.8036	0.0000	Yes
2	rv_sp500	rv_wti	5.7388	0.0001	Yes
3	rv_sp500	rv_hy	12.0214	0.0000	Yes
4	rv_hy	rv_wti	7.0133	0.0000	Yes
5	rv_hy	rv_sp500	5.4570	0.0002	Yes

Impulse Response Functions

```
In [58]: #impulse responses with Cholesky ordering: WTI, HY, SP500
variables_var1 = ["ret_wti", "d_spread_hy", "er_sp500"]
variables_var2 = ["rv_wti", "rv_hy", "rv_sp500"]

irf_var1_fitted = VAR(df_final[variables_var1]).fit(2)
irf_var2_fitted = VAR(df_final[variables_var2]).fit(4)

def compute_irfs_shock(lag_matrices, cholesky_P, shock_vector, n_periods)
    n_lags = len(lag_matrices)
    n_vars = lag_matrices.shape[1]

    responses = np.zeros((n_periods + 1, n_vars))
    responses[0] = cholesky_P @ shock_vector

    for period in range(1, n_periods + 1):
        for lag in range(1, min(period, n_lags) + 1):
            responses[period] += lag_matrices[lag-1] @ responses[period - lag]

    return responses

oil_shock_vector = np.array([1, 0, 0])

# VAR 1 (returns)
P1 = np.linalg.cholesky(irf_var1_fitted.sigma_u)
manual1_shock_matrix = compute_irfs_shock(irf_var1_fitted.coefs, P1, oil_shock_vector)
print(f"VAR 1 : t=0 Impulse vector (from WTI return shock): {np.round(manual1_shock_matrix, 2)}")

# VAR 2 (realized variances)
P2 = np.linalg.cholesky(irf_var2_fitted.sigma_u)
manual2_shock_matrix = compute_irfs_shock(irf_var2_fitted.coefs, P2, oil_shock_vector)
print(f"VAR 2 : t=0 Impulse vector (from WTI variance shock): {np.round(manual2_shock_matrix, 2)}")

# VAR 1 plots
fig, axes = plt.subplots(1, 3, figsize=(18, 4))
for i, (ax, title) in enumerate(zip(axes, variables_var1)):
    ax.plot(manual1_shock_matrix[:, i], color="steelblue", marker='o', markersize=10)
    ax.axhline(0, color="black", linewidth=0.8)
    ax.grid(alpha=0.3)
    ax.set_xticks(range(13))
    ax.set_title(f"Response of {title}")
```

```

ax.set_xlabel("Weeks")
if i == 0:
    ax.set_ylabel("Impact")
fig.suptitle("VAR 1 : System Response to a 1 s.d. WTI return shock", y=1.
plt.tight_layout()
plots_dir = Path("../plots")
fig.savefig(plots_dir/ "irf_var1.png", dpi=300, bbox_inches="tight")
plt.show()

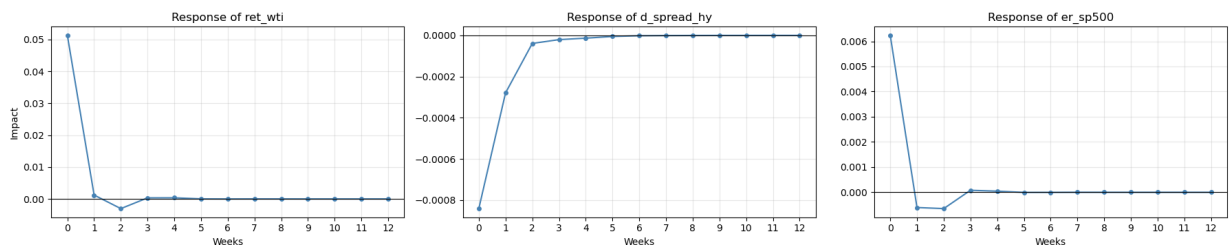
# VAR 2 plots
fig, axes = plt.subplots(1, 3, figsize=(18, 4))
for i, (ax, title) in enumerate(zip(axes, variables_var2)):
    ax.plot(manual2_shock_matrix[:, i], color="steelblue", marker='o', ma
    ax.axhline(0, color="black", linewidth=0.8)
    ax.grid(alpha=0.3)
    ax.set_xticks(range(13))
    ax.set_title(f"Response of {title}")
    ax.set_xlabel("Weeks")
    if i == 0:
        ax.set_ylabel("Impact")
fig.suptitle("VAR 2 : System Response to a 1 s.d. WTI variance shock", y=
plt.tight_layout()
fig.savefig(plots_dir/ "irf_var2.png", dpi=300, bbox_inches="tight")
plt.show()

```

VAR 1 : t=0 Impulse vector (from WTI return shock): [0.051284 -0.0008
42 0.006236]

VAR 2 : t=0 Impulse vector (from WTI variance shock): [2.527e-03 2.000e-0
6 1.570e-04]

VAR 1 : System Response to a 1 s.d. WTI return shock



VAR 2 : System Response to a 1 s.d. WTI variance shock

